

PRACTICAL -1

Task 1: Develop a C program that demonstrates how a Linux BTL operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child 4 process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait O. 5. Display the Process ID (PID) of both parent and child processes.

```
ruhi0404@Ruhitha:~/Operating_Systems$ nano skill1task2.c
ruhi0404@Ruhitha:~/Operating_Systems$ nano pracltask1.c
ruhi0404@Ruhitha:~/Operating_Systems$ gcc pracltask1.c -o pracltask1
ruhi0404@Ruhitha:~/Operating_Systems$ ./pracltask1
Enter a Linux command: ls
Parent Process: PID = 586
Child PID = 587
Child Process: PID = 587
Parent PID = 586
File1.txt  Makefile  bin      obj      pra3task1  pra3task2.c  practical  skill1task2  skill2task1.c  skill2w1task1  skill2w1task2.c  src
File2.txt  README.md  docs     praltask1  pra3task1.c  pracltask1  scripts   skill1task2.c  skill2task2    skill2w1task1.c  skillw3task2     tests
LICENSE    assets     include  praltask1.c  pra3task2    pracltask1.c  skill    skill2task1    skill2task2.c  skill2w1task2    skillw3task2.c
Child process completed.
ruhi0404@Ruhitha:~/Operating_Systems$ pwd
/home/ruhi0404/Operating_Systems
ruhi0404@Ruhitha:~/Operating_Systems$ date
Tue Aug 11 06:53:53 UTC 2026
ruhi0404@Ruhitha:~/Operating_Systems$ |
```

Observation: The program successfully accepts a Linux command from the user and creates a child process using fork(). The child process executes the entered command using execlp(), while the parent process waits for the child using wait(). The PIDs of both the parent and child processes are displayed, demonstrating process creation, command execution, and parent-child process synchronization in Linux.

Task 2: Using Linux terminal commands (uname, Iscpu, Isblk, PS, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining 4 how the OS abstracts CPU , memory, storage, and 1/0 devices.

```
ruhi0404@Ruhitha:~/Operating_Systems$ uname -a
Linux Ruhitha 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux
ruhi0404@Ruhitha:~/Operating_Systems$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 8
On-line CPU(s) list:    0-7
Vendor ID:              GenuineIntel
Model name:             Intel(R) Core(TM) i7-10610U CPU @ 1.80GHz
CPU family:             6
Model:                 142
Thread(s) per core:     2
Core(s) per socket:     4
Socket(s):              1
Stepping:               12
BogoMIPS:               4608.01
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtsc
cp_l1m arch_perfmon rep_good nopl xtopology cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pdcm pcid sse4_1 sse4_2 movbe
popcnt aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch srbds ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpid ep
t_ad fsgsbase bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt xsaveopt xsavec xgetbv1 xsaves vnmi md_clear flush_l1
d arch_capabilities

Virtualization features:
Virtualization:         VT-x
Hypervisor vendor:      Microsoft
Virtualization type:    full
Caches (sum of all):
L1d:                    128 KiB (4 instances)
L1i:                    128 KiB (4 instances)
L2:                      1 MiB (4 instances)
L3:                      8 MiB (1 instance)
NUMA:
NUMA node(s):           1
NUMA node0 CPU(s):      0-7
Vulnerabilities:
Gather data sampling:    Not affected
Ghostwrite:              Not affected
Indirect target selection: Mitigation; Aligned branch/return thunks
Itlb multihit:           KVM: Mitigation: Split huge pages
L1tf:                    Not affected
```

```
Reg file data sampling: Not affected
Retbleed:                Mitigation; Enhanced IBRS
Spec rstack overflow:    Not affected
Spec store bypass:       Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1:              Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2:              Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHT SW loop, KVM SW loop
Sbds:                    Unknown; Dependent on hypervisor status
Tsx:                     Not affected
Tsx async abort:         Mitigation; TSX disabled
Vmscape:                 Not affected
ruhi0404@Ruhitha:~/Operating_Systems$
ruhi0404@Ruhitha:~/Operating_Systems$
lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda 8:0 0 356.9M 1 disk
sdb 8:16 0 159.4M 1 disk
sdc 8:32 0 4G 0 disk [SWAP]
sdd 8:48 0 1T 0 disk /mnt/wslg/distro

ruhi0404@Ruhitha:~/Operating_Systems$ ps
PID TTY TIME CMD
608 pts/0 00:00:00 bash
ruhi0404@Ruhitha:~/Operating_Systems$ ps aux
USER PID %CPU %MEM VSZ RSS TTY STAT START TIME COMMAND
root 1 0.1 0.0 21724 13204 ? Ss 06:30 0:02 /sbin/init
root 2 0.0 0.0 3180 2204 hvc0 Sl+ 06:30 0:00 /init
root 7 0.0 0.0 3180 2040 hvc0 Sl+ 06:30 0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root 46 0.0 0.0 50368 16008 ? Ss 06:30 0:00 /usr/lib/systemd/systemd-journald
root 96 0.0 0.0 25020 6336 ? Ss 06:30 0:01 /usr/lib/systemd/systemd-udevd
systemd+ 123 0.0 0.0 21344 13244 ? Ss 06:30 0:00 /usr/lib/systemd/systemd-resolved
systemd+ 131 0.0 0.0 91036 7988 ? Ssl 06:30 0:00 /usr/lib/systemd/systemd-timesyncd
root 170 0.0 0.0 4244 2688 ? Ss 06:30 0:00 /usr/sbin/cron -f -p
message+ 171 0.0 0.0 9636 5372 ? Ss 06:30 0:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --sysl
root 181 0.0 0.0 17976 8788 ? Ss 06:30 0:00 /usr/lib/systemd/systemd-logind
syslog 201 0.0 0.0 222516 5680 ? Ssl 06:30 0:00 /usr/sbin/rsyslogd -n -iNONE
root 204 0.0 0.0 3124 1972 tty1 S+ 06:30 0:00 /sbin/agetty -o -p -- \u --noclear - linux
root 205 0.0 0.1 107036 23172 ? Ssl 06:30 0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-si
root 206 0.0 0.0 3184 1108 ? Ss 06:30 0:00 /init
root 297 0.0 0.0 3200 1244 ? S 06:30 0:00 /init
```

```
root 204 0.0 0.0 3124 1972 tty1 Ss+ 06:30 0:00 /sbin/agetty -o -p -- \u --noclear - linux
root 205 0.0 0.1 107036 23172 ? Ssl 06:30 0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-si
root 296 0.0 0.0 3184 1108 ? Ss 06:30 0:00 /init
root 297 0.0 0.0 3200 1244 ? S 06:30 0:00 /init
ruhi0404 302 0.0 0.0 6076 5380 pts/0 Ss 06:30 0:00 -bash
root 305 0.0 0.0 6704 4676 pts/1 Ss 06:30 0:00 /bin/login -f
ruhi0404 364 0.0 0.0 20324 11508 ? Ss 06:30 0:00 /usr/lib/systemd/systemd --user
ruhi0404 366 0.0 0.0 21164 3572 ? S 06:30 0:00 (sd-pam)
ruhi0404 390 0.0 0.0 6080 5372 pts/1 S+ 06:30 0:00 -bash
ruhi0404 609 0.0 0.0 8288 4300 pts/0 R+ 06:57 0:00 ps aux
ruhi0404@Ruhitha:~/Operating_Systems$ top
top - 06:57:21 up 30 min, 1 user, load average: 0.00, 0.01, 0.00
Tasks: 21 total, 1 running, 20 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 99.9 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 15853.6 total, 15341.8 free, 554.9 used, 152.8 buff/cache
MiB Swap: 4096.0 total, 4096.0 free, 0.0 used, 15298.7 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	21724	13204	9796	S	0.0	0.1	0:02.32	systemd
2	root	20	0	3180	2204	2072	S	0.0	0.0	0:00.03	init-systemd(Ub
7	root	20	0	3180	2040	1940	S	0.0	0.0	0:00.00	init
46	root	19	-1	50368	16008	14852	S	0.0	0.1	0:00.73	systemd-journal
96	root	20	0	25020	6336	5156	S	0.0	0.0	0:01.02	systemd-udevd
123	systemd+	20	0	21344	13244	11116	S	0.0	0.1	0:00.42	systemd-resolve
131	systemd+	20	0	91036	7988	7024	S	0.0	0.0	0:00.33	systemd-timesyn
170	root	20	0	4244	2688	2432	S	0.0	0.0	0:00.02	cron
171	message+	20	0	9636	5372	4668	S	0.0	0.0	0:00.32	dbus-daemon
181	root	20	0	17976	8788	7748	S	0.0	0.1	0:00.30	systemd-logind
201	syslog	20	0	222516	5680	4660	S	0.0	0.0	0:00.20	rsyslogd
204	root	20	0	3124	1972	1836	S	0.0	0.0	0:00.01	agetty
205	root	20	0	107036	23172	13716	S	0.0	0.1	0:00.35	unattended-upgr
206	root	20	0	3184	1108	980	S	0.0	0.0	0:00.00	SessionLeader
297	root	20	0	3200	1244	1108	S	0.0	0.0	0:00.22	Relay(302)
302	ruhi0404	20	0	6076	5380	3656	S	0.0	0.0	0:00.20	bash
305	root	20	0	6704	4676	3888	S	0.0	0.0	0:00.03	login
364	ruhi0404	20	0	20324	11508	9384	S	0.0	0.1	0:00.38	systemd
366	ruhi0404	20	0	21164	3572	1848	S	0.0	0.0	0:00.00	(sd-pam)
390	ruhi0404	20	0	6080	5372	3704	S	0.0	0.0	0:00.09	bash
610	ruhi0404	20	0	9300	5680	3492	R	0.0	0.0	0:00.01	top

Observation

The Linux terminal commands were used to investigate the relationship between hardware resources and operating system services. The `uname` command provided information about the operating system and kernel. The `lscpu` command displayed CPU architecture, processor count, and CPU characteristics. The `lsblk` command displayed the available block storage devices and their partitions. The `ps` command displayed running processes, while `top` provided real-time information about processes, CPU usage, and memory usage.

How the OS abstracts hardware resources

1. CPU:

The operating system abstracts the CPU by providing processes and threads with CPU time without requiring applications to directly control the processor. The scheduler decides which process runs and for how long.

2. Memory:

The OS provides each process with its own virtual address space. Applications can use memory without needing to know the physical RAM locations. The OS manages allocation, protection, and virtual memory.

3. Storage:

The OS abstracts physical storage devices through files and directories. Applications can read and write files without directly controlling disk sectors. File systems and device drivers handle the underlying storage operations.

4. I/O Devices:

Input/output devices such as keyboards, displays, network cards, and USB devices are accessed through operating-system interfaces and device drivers. Applications do not normally need to communicate directly with the hardware.